



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. CHEMISTRY				
Course Title	THEORETICAL CHEMISTRY III - ADVANCED QUANTUM CHEMISTRY				
Type of Course	MAJOR				
Semester	VII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	A good understanding of the concepts learned in the course, CHE3CJ201 - Theoretical Chemistry 1 – Basics of Quantum Chemistry - Postulates of quantum mechanics and related concepts, Application of these concepts to particle in a 1D box and 3D box.				
Course Summary	In the course, CHE3CJ201 , students learned the basic concepts of Quantum Chemistry including the postulates of quantum mechanics and also learned how to apply these concepts to different systems. This course further develops the concepts by applying them to other systems of chemical relevance. While doing so, the subject matter can be related to other application level topics like Molecular Spectroscopy and Group Theory. Students also realize that for systems having more than one electrons (atoms other than hydrogen and molecules) time-independent Schrodinger equation could not be exactly solved. To overcome this difficulty, excellent approximate techniques are formulated and they can be employed with the help of computer programmes for solving any system of chemical relevance. Thus students are equipped with the basic concepts and methods of a very significant research area called Computational Chemistry. Hands on experience in computational chemistry can be gained through the practical sessions given in module V.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	<i>recall</i> the quantum mechanical postulates as the fundamental principles of quantum chemistry.	R	F	Quiz/MCQ Test

CO2	<i>realize</i> the wave functions of hydrogen atom as atomic orbitals.	U	C	Assignment/Viva
CO3	<i>apply</i> 1D Simple Harmonic Oscillator as a preliminary model for deriving the quantised energy levels of normal modes of vibration in molecules.	Ap	P	Class Test/Problem solving sessions
CO4	<i>relate</i> particle on a sphere as a starting model for deducing the quantised energy levels of rotation in diatomic molecules.	An	P	Class Test/Problem solving sessions
CO5	<i>justify</i> the use of approximate methods for deducing the wave functions and energy values of multi-electron systems.	E	P	Class Test/Problem solving sessions
CO6	<i>propose</i> computational methods for solving real world problems in chemistry	C	M	Assignment/Seminar
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (40+30)	Marks
I	Quantum Mechanics of Vibrational and Rotational Motions		15	31
	1	Review of the postulates of quantum mechanics.	1	
	2	One-dimensional simple harmonic oscillator (complete treatment):- Method of power series, Hermite equation and Hermite polynomials, recursion relation, wave functions and their plots, energy eigenvalues, important features of the problem, harmonic oscillator model and molecular vibrations.	3	
	3	Cartesian and spherical polar coordinates and their relationships.	1	
	4	Planar rigid rotor (or particle on a ring), the Phi-equation, solution of the Phi-equation.	2	
	5	One particle Rigid rotator (non-planar rigid rotator or particle on a sphere) (complete treatment): The wave	3	

		equation in spherical polar coordinates, separation of variables, the Phi-equation and the Theta-equation and their solutions, Legendre and Associated Legendre equations, Legendre and Associated Legendre polynomials.		
	6	Spherical harmonics (imaginary and real forms), polar diagrams of spherical harmonics.	2	
	7	Quantization of angular momentum, space quantization, quantum mechanical operators corresponding to angular momenta ($\hat{L}_x, \hat{L}_y, \hat{L}_z, \hat{L}^2$), commutation relations between these operators, Ladder operator method for angular momentum.	3	
	Sections from References: Section A			
II	Quantum Mechanics of Hydrogen-like Atoms		10	25
	8	Potential energy of hydrogen-like systems, the wave equation in spherical polar coordinates, separation of variables, the R, Theta and Phi equations and their solutions, Laguerre and associated Laguerre polynomials, atomic orbitals/wave functions of hydrogen-like atoms and their corresponding energy.	4	
	9	Radial and angular parts of atomic orbitals - Radial functions and radial plots, radial distribution functions and their plots, angular functions (spherical harmonics) and their plots.	2	
	10	The postulate of spin by Uhlenbeck and Goudsmith - Spin orbitals, construction of spin orbitals from orbitals and spin functions.	2	
	11	Pauli's principle of anti-symmetric wave functions - Slater determinants.	2	
	Sections from References: Section A			
III	Approximation Methods in Quantum Mechanics		12	28
	12	Many body problem and the need of approximation methods;	1	
	13	Independent particle model – Application to the ground state of helium atom.	1	
	14	Variation method – Variation theorem with proof, illustration of variation theorem using the trial function x	3	

		(a-x) for particle in a 1D-box, variation treatment of the ground state of helium atom.		
	15	Perturbation method – Time-independent perturbation method (non-degenerate case only), perturbation treatment of the ground state of helium atom.	3	
	16	Hartree's Self-Consistent Field method for atoms, Fock modification using spin orbitals & Hartree -Fock Self-Consistent Field (HF-SCF) method for atoms, the Fock operator;	4	
	Sections from References: Section A			
IV	Introduction to Computational Chemistry		8	14
	17	Classification of Computational Chemistry methods – Molecular mechanics methods (the concept of the force field) and Electronic structure methods, ab initio and semi-empirical methods (Basic idea only).	2	
	18	Roothan's concept of basis functions - Slater type orbitals (STO) and Gaussian type orbitals (GTO).	1	
	19	Concept of electron correlation and post HF methods. (Elementary idea)	1	
	20	Basis set approximation in ab initio methods - classification of basis sets – minimal, double zeta, triple zeta, split-valence, polarization & diffuse basis sets, Pople-style basis sets, and their nomenclature.	2	
	21	Gaussian programme – The structure of a Gaussian input file, Types of keywords.	1	
	22	Specification of molecular geometry using a) Cartesian coordinates and b) Internal coordinates. The Z-matrix, Z-matrices of some simple molecules like H ₂ , H ₂ O, HCHO and NH ₃ .	1	
	Sections from References: Section B			
V	Computational Chemistry Practical		30	
	1	1. Single point energy calculations of simple molecules like H ₂ O and NH ₃ at the HF/3-21G level of theory. 2. The effect of basis set on the single point energy of H ₂ O and NH ₃ using the Hartree-Fock method (3-21G, 6-31G, 6-31+G, 6-31+G* basis sets can be used).	10	

		3. Geometry optimization of molecules like H ₂ O, NH ₃ , HCHO & C ₂ H ₄ at the HF/6-31G level of theory. 4. Computation of dipole and quadrupole moments of HCHO & C ₂ H ₄ at the HF/6-31G level of theory.		
	2	5. Effect of basis set on the computation of H-O-H bond angle in H ₂ O using the HartreeFock method (3-21G, 6-31G, 6-31+G, 6-31+G* basis sets can be used). 6. Computation of the energy of HOMO and LUMO of formaldehyde and ethylene at the HF/6-31G level of theory. 7. Effect of substituent (F & Cl) on the geometric parameters (like C-C bond length) of ethylene at the HF/6-31G level of theory.	8	
	3	8. Comparison of stability of cis-planar and trans-planar conformers of H ₂ O ₂ at the HF/6-31G level of theory. 9. Comparison of stability of cis- and trans-isomers of difluoroethylene at the HF/6-31G* level of theory.	6	
	Open - ended	• Determination of hydrogen bond strength of H₂O dimer and H₂O trimer at the HF/6-31+G* level of theory. • Computation of the frequencies of normal modes of vibration of molecules like H₂O, NH₃ and CO₂ at the HF/6-31+G* level of theory.	6	
	Sections from References: Section C			

Books and References:

Section A

1. I. N. Levine, *Quantum Chemistry*, 6th Edn., Pearson Education Inc., 2009.
2. P. W. Atkins, R. S. Friedman, *Molecular Quantum Mechanics*, 4th Edn., Oxford University Press, 2005
3. Donald, A. McQuarrie, *Quantum Chemistry*, University Science Books, 1983 (first Indian edition, Viva books, 2003).
4. R.K. Prasad, *Quantum Chemistry*, 3rd Edition, New Age International, 2006

Section B

1. Frank Jensen, Introduction to Computational Chemistry, John Wiley & Sons, 1999.
2. C. J. Cramer, Essentials of computational Chemistry: Theories and models, John Wiley & Sons 2002.

Section C

1. J. Foresman & Aelieen Frisch, Exploring Chemistry with Electronic Structure Methods, Gaussian Inc., 2000.
2. David Young, Computational Chemistry - A Practical Guide for Applying Techniques to Real-World Problems”, Wiley-Interscience, 2001.

Further reading

1. F.L. Pilar, Elementary Quantum Chemistry, McGraw-Hill, 1968
2. Thomas Engel, Quantum Chemistry & Spectroscopy, Pearson Education, 2006
3. Errol G. Lewars, Computational Chemistry: Introduction to the theory and applications of molecular quantum mechanics, 2nd edn., Springer 2011.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1	1	-	-	-	2	3				2		2
CO 2	2	3	-	-	-	2	3				1		2
CO 3	-	2		-	-	2	3		2	2	3	1	2
CO 4	-	2			-	2	3		1	2	3		3
CO 5	-	2	-	-	-	3	3				2	2	3
CO 6	-	3	-		-	3	3		2	2	3	2	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment/viva/seminar	Practical skill Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2		✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓		